

MA206 AY27-1 WPR I Review

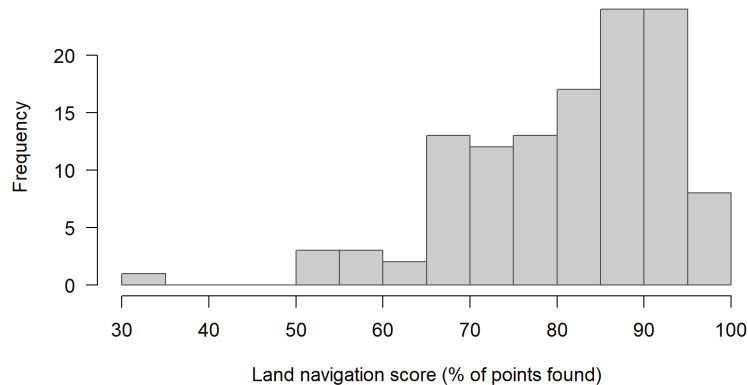
Descriptive Statistics, Probability, and Random Variables

On the WPR you are authorized the SRC, R-Lite, and your issued calculator. Give final answers to 3 decimal places.

Part A: Concept Checks

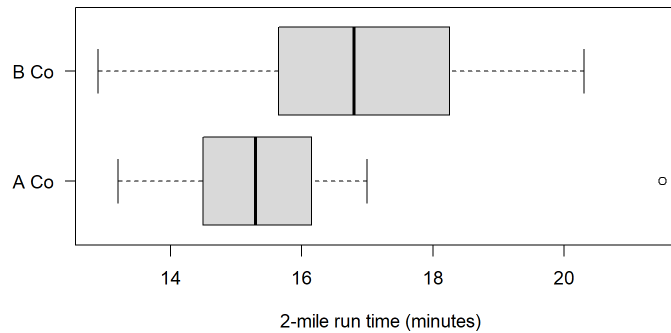
True/False, multiple choice, or fill in the blank. For every "False," be able to say why.

1. **True or False:** If $P(A) = 0.65$ and $P(B) = 0.55$, then A and B could be mutually exclusive.
2. An analyst records the blood type (A, B, AB, O) of 40 Soldiers. This variable is:
(a) categorical (nominal) (b) categorical (ordinal)
(c) numerical, discrete (d) numerical, continuous
3. **True or False:** A cumulative distribution function $F(x)$ can decrease as x increases.
4. Classify each study. (i) A researcher compares the GPAs of cadets who *choose* to drink energy drinks with those who do not. (ii) A PT instructor randomly assigns platoons to one of two stretching routines and compares injury rates.
(a) (i) observational study; (ii) designed experiment
(b) (i) designed experiment; (ii) observational study
(c) both are simple random samples
(d) both are censuses
5. A die is rolled. Let A = "the roll is even" and B = "the roll is greater than 3." The event $A \cap B'$ is:
(a) $\{2\}$ (b) $\{2, 4, 6\}$
(c) $\{4, 6\}$ (d) $\{1, 3\}$
6. **Fill in the blanks:** If $X \sim \text{Pois}(\mu = 6)$, then $E(X) = \underline{\hspace{1cm}}$, $V(X) = \underline{\hspace{1cm}}$, and $\text{SD}(X) = \underline{\hspace{1cm}}$.
7. **True or False:** If the largest value in a sample is changed to a much larger value, the sample mean can change a great deal while the sample median does not change at all.
8. **True or False:** For any events A and B , $(A \cup B)' = A' \cap B'$.
9. **True or False:** For a continuous random variable, $f(3)$ is the probability that $X = 3$.
10. You roll a six-sided die and flip a coin. How many outcomes are in the sample space?
(a) 6 (b) 8
(c) 12 (d) 36
11. The histogram below shows land navigation scores for 120 cadets. The shape is best described as:
(a) right-skewed (b) symmetric
(c) left-skewed (d) uniform



12. For the histogram in the previous question, which relationship is most likely?
 - (a) mean > median
 - (b) mean $\approx$ median
 - (c) mean < median
 - (d) cannot tell
13. If $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cup B) = 0.7$, then $P(A \cap B)$ is:
 - (a) 0.2
 - (b) 0.9
 - (c) 0.3
 - (d) 0.1
14. **Fill in the blank (memoryless property):** If T is exponential, $P(T > 5 \mid T > 2) = P(T > \text{_____})$.
15. **True or False:** The average fuel consumption of *every* vehicle in a brigade last month is a statistic.
16. **True or False:** A squad of 11 Soldiers includes 4 medics. Select 5 Soldiers *without replacement*. Then X = the number of medics selected is binomial.
17. **Fill in the blank:** If every value in a data set is multiplied by 3, the new sample variance is _____ times the original sample variance.
18. **Fill in the blank:** $P(A') = \text{_____}$.
19. A screening test detects a condition 95% of the time when it is present and gives a false alarm 5% of the time when it is absent. 1% of Soldiers have the condition. A Soldier tests positive. The probability they have the condition is closest to:
 - (a) 0.95
 - (b) 0.50
 - (c) 0.16
 - (d) 0.01
20. **Fill in the blank:** If X is a continuous random variable, $P(X = 2.5) = \text{_____}$, so $P(X \leq 2.5)$ and $P(X < 2.5)$ are _____.
21. The critical value $z_{0.05} = 1.645$ means:
 - (a) $P(Z < 1.645) = 0.05$
 - (b) $P(Z > 1.645) = 0.05$
 - (c) $P(|Z| > 1.645) = 0.05$
 - (d) $P(Z = 1.645) = 0.05$
22. A plant tests batteries from a production line that will keep producing batteries next year. The collection of all batteries of interest is best described as:
 - (a) a finite population
 - (b) a process (conceptual population)
 - (c) a sample
 - (d) a parameter

23. **True or False:** The Poisson distribution models the *time* between arrivals.
24. **True or False:** There are more ways to choose 3 cadets from 10 than to choose 7 cadets from 10.
25. **True or False:** If 10 minutes is added to every run time in a data set, the standard deviation increases by 10.
26. Which of the following is a valid probability mass function on $x = 0, 1, 2$?
 (a) 0.3, 0.3, 0.3 (b) 0.5, 0.6, -0.1 (c) 0.25, 0.50, 0.25 (d) 0.4, 0.4, 0.4
27. $P(A) = 0.3$ and $P(B) = 0.5$. The events A and B are independent exactly when $P(A \cap B)$ equals:
 (a) 0.8 (b) 0.15
 (c) 0.2 (d) 0
28. The comparative boxplots show 2-mile run times for two companies.



- (i) Which company has the larger median time? (ii) Which has the larger IQR?
 (iii) Does either company show an outlier?
29. The number of push-ups a Soldier completes in two minutes is:
 (a) categorical (nominal) (b) categorical (ordinal)
 (c) numerical, discrete (d) numerical, continuous
30. **True or False:** For the standard normal random variable Z , $P(Z < -1.5) = P(Z > 1.5)$.
31. **True or False:** Two mutually exclusive events, each with positive probability, are independent.
32. Which of the following is best modeled with a Poisson distribution?
 (a) the number of heads in 20 coin flips
 (b) the number of flat tires per 100 miles of convoy travel, at a steady average rate
 (c) the time until the next flat tire
 (d) the weight of a tire
33. **True or False:** A probability density function $f(x)$ can take values greater than 1.
34. **True or False:** In a simple random sample of size n , every possible sample of size n has the same chance of being selected.

35. **Fill in the blanks:** If $X \sim \text{Bin}(n, p)$, then $E(X) = \underline{\hspace{2cm}}$ and $V(X) = \underline{\hspace{2cm}}$.
36. **True or False:** Surveying the first 30 cadets who walk into the mess hall produces a simple random sample.
37. **True or False:** If A and B are independent, then A' and B are also independent.
38. A pdf that is *decreasing* across its entire support describes a distribution that is:
(a) left-skewed (b) symmetric
(c) right-skewed (d) bimodal
39. A company commander summarizes her company's ACFT scores with a mean and a boxplot and draws no conclusions beyond her company. This is:
(a) descriptive statistics (b) inferential statistics
(c) a designed experiment (d) a census of the Army
40. If $T \sim \text{Exp}(\lambda = 0.08)$, the median of T is:
(a) 12.5 (b) 8.664
(c) 0.08 (d) 6.25
41. **Fill in the blanks:** $E(aX + b) = \underline{\hspace{2cm}}$ and $V(aX + b) = \underline{\hspace{2cm}}$.
42. **True or False:** If $T \sim \text{Exp}(\lambda = 0.5)$, then the standard deviation of T is 0.5.
43. **True or False:** For any events A and B , $P(A \mid B) = P(B \mid A)$.
44. Which of the following is *not* a requirement for a binomial experiment?
(a) a fixed number of trials n (b) independent trials (c) the same p on every trial
(d) the success probability must be $p = 0.5$

Part B: Quick Computations

Short calculations. Topics are mixed.

1. A vehicle bumper code is 3 letters followed by 2 digits, and repeats are allowed. How many codes are possible?

The next three questions use these 9 observations: 3, 21, 22, 24, 25, 26, 27, 29, 38.
Find Q_1 and Q_3 as the medians of the lower and upper halves, including the median in both halves when n is odd (Devore's fourths).

2. Find the sample mean and the sample median.
3. Find Q_1 , Q_3 , and the IQR.
4. Use the $1.5 \times$ IQR rule to identify any outliers.
5. $X \sim N(\mu = 50, \sigma = 8)$. Find $P(X < 62)$.
6. $X \sim N(\mu = 50, \sigma = 8)$. Find the 95th percentile of X .
7. $P(A) = 0.5$, $P(B) = 0.3$, and $P(A \cap B) = 0.1$. Draw a Venn diagram and label the probability of all four regions. Then find $P(A' \cap B')$ and $P(A \cap B')$.
8. The pdf of X is $f(x) = \frac{2(3-x)}{9}$ for $0 \leq x \leq 3$ (and 0 otherwise). Find $P(-1 \leq X \leq 1)$.
9. Four components work independently, each with probability 0.95. Find the probability all four work.

The next question uses this pmf:

x	0	1	2	3
$p(x)$	0.1	0.3	0.4	0.2

10. Find $E(X)$ and $V(X)$.
11. Find the mean and the 10% trimmed mean of: 2, 14, 15, 16, 16, 17, 18, 19, 20, 22.
12. The wait for a shuttle is exponential with a mean of 25 minutes. Find the probability of waiting more than 30 minutes.
13. A briefer must present 3 of 10 topics, and the order of presentation matters. How many briefing orders are possible?
14. $X \sim \text{Bin}(8, 0.2)$. Find $P(X = 2)$.
15. $X \sim \text{Bin}(8, 0.2)$. Find $P(X \geq 1)$.
16. Find the sample variance, sample standard deviation, and range of: 4, 7, 9, 10, 15.
17. Aircraft land at an airfield at an average rate of 1.5 per hour. Find the probability of *no* landings in a 3-hour window.
18. $P(A \cap B) = 0.12$ and $P(B) = 0.40$. Find $P(A | B)$.

19. A section has 5 women and 7 men. Four cadets are chosen at random for a briefing team. What is the probability exactly 2 are women?
20. In a battalion, 30% of Soldiers are Airborne qualified, 40% Air Assault qualified, and 10% Ranger qualified. 15% hold both Airborne and Air Assault, 8% Airborne and Ranger, 7% Air Assault and Ranger, and 5% hold all three. Find the probability a randomly chosen Soldier holds at least one of the three.

Part C: Full Problems

Define your random variable before computing.

Problem 1. Let X be the number of cadets in a squad who are late to morning formation, with pmf

$$p(x) = \begin{cases} 0.10 & x = 0 \\ 0.25 & x = 1 \\ 0.30 & x = 2 \\ 0.20 & x = 3 \\ 0.15 & x = 4 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the cdf $F(x) = P(X \leq x)$ and write it in full piecewise form.
- (b) Find $P(X < 2)$.
- (c) Find $P(X \leq 2)$.
- (d) Find $P(X > 3)$.
- (e) Find $P(X \geq 3)$.
- (f) Find μ and σ . Then find the probability X is within one standard deviation of its mean.

Problem 2. The time X (in hours) a cadet spends on homework in an evening has pdf

$$f(x) = \begin{cases} \frac{x}{8} & 0 \leq x \leq 4 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the cdf $F(x) = P(X \leq x)$ and write it in full piecewise form.
- (b) Find $P(X < 2)$.
- (c) Find $P(X \leq 2)$.
- (d) Find $P(X > 3)$.
- (e) Find $P(X \geq 3)$.
- (f) Find $P(X = 3)$.
- (g) Find μ and σ . Then find the probability X is more than one standard deviation above its mean.

Problem 3. Two-mile run times X for a company are normal with mean 15.5 minutes and standard deviation 1.2 minutes.

- (a) Find $P(X < 14)$.
- (b) Find $P(X \leq 14)$.
- (c) Find $P(X > 18)$.
- (d) Find $P(X \geq 18)$.
- (e) Find $P(X = 18)$.
- (f) Find μ and σ . Then find the probability X is more than two standard deviations above its mean.

Problem 4. Each of 15 Soldiers independently qualifies Expert at the range with probability 0.3. Let X be the number who qualify Expert, so $X \sim \text{Bin}(15, 0.3)$.

- (a) Find $P(X < 4)$.
- (b) Find $P(X \leq 4)$.
- (c) Find $P(X > 6)$.
- (d) Find $P(X \geq 6)$.
- (e) Find μ and σ . Then find the probability X is more than two standard deviations below its mean.

Problem 5. Calls to the staff duty desk arrive randomly and independently at an average rate of 6 per hour. Let X be the number of calls in one hour.

- (a) What distribution does X follow?
- (b) Find $P(X < 5)$.
- (c) Find $P(X \leq 5)$.
- (d) Find $P(X > 8)$.
- (e) Find $P(X \geq 8)$.
- (f) Find μ and σ . Then find the probability X is within two standard deviations of its mean.

Problem 6. Return to the staff duty desk in Problem 5, with calls arriving at an average rate of 6 per hour. Now let T be the time (in minutes) between one call and the next.

- (a) What distribution does T follow? Write its pdf. Then find the cdf $F(t) = P(T \leq t)$ and write it in full piecewise form.
- (b) Find $P(T < 6)$.

- (c) Find $P(T \leq 6)$.
- (d) Find $P(T > 15)$.
- (e) Find $P(T \geq 15)$.
- (f) Find $P(T = 15)$.
- (g) Find μ and σ . Then find the probability T is within one standard deviation of its mean.

Problem 7. In a motor pool, 60% of vehicles are wheeled and 40% are tracked. 10% of wheeled vehicles and 25% of tracked vehicles fail their monthly inspection. A vehicle is selected at random.

- (a) Find the probability the vehicle fails inspection.
- (b) The vehicle failed inspection. What is the probability it is tracked?